



Generative artificial intelligence and happiness as drivers of competitiveness and sustainable development in green hotel environments

Journal:	<i>International Journal of Productivity and Performance Management</i>
Manuscript ID	IJPPM-04-2025-0303.R2
Manuscript Type:	Standard Paper
Keywords:	Green hotels, happiness, Sustainability, generative artificial intelligence, Technology

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Table 1. Indicators

Constructs	Items	Authors
Management Process s (MP)	MP1: Do you consider the sustainability practices implemented at Green Hotels to be effective?	(Acampora, Lucchetti, et al., 2022; Chen et al., 2022; Merli et al., 2019)
	MP2: Do you consider the use of sustainable building materials and furnishings in Green Hotels to be appropriate?	(Acampora, Lucchetti, et al., 2022; M. F. Ahmed et al., 2021)
	MP3: To what extent do you perceive senior management's commitment to green leadership in Green Hotels?	(Haldorai et al., 2022a; Munawar et al., 2022)
	MP4: To what extent do you perceive the collaboration of Green Hotels with the local community for sustainable development?	(Chen et al., 2022; Merli et al., 2019; Wei et al., 2023)
Environmental Performance(EP)	EP1: To what extent does the company value and promote environmental responsibility among its employees?	(Abuelhassan & Elsayed, 2020; Raza & Khan, 2022; Yong et al., 2022)
	EP2: To what extent does the organisational culture of the company encourage the adoption of sustainable practices at work?	(Abuelhassan & Elsayed, 2020; Raza & Khan, 2022; Yong et al., 2022)
	EP3: To what extent does the Green Hotels' commitment to environmental protection manifest itself in their policies and actions?	(Abuelhassan & Elsayed, 2020; Raza & Khan, 2022; Yong et al., 2022)
	EP4: To what extent do you consider that the company's practices and values are aligned with the principles of environmental responsibility?	(Abuelhassan & Elsayed, 2020; Raza & Khan, 2022; Yong et al., 2022)
	EP5: To what extent does organisational culture influence your willingness to participate in sustainability activities at work?	(Abuelhassan & Elsayed, 2020; Raza & Khan, 2022; Yong et al., 2022)
Awareness (AG)	AG 1: To what extent do you see Green Hotels as promoting global environmental awareness?	(Cembruch-Nowakowski, 2019)

	AG 2: To what extent do you consider that training and skills acquisition in sustainability leads to greater ecological commitment?	(Alreahi et al., 2023; Pham et al., 2018)
	AG 3: To what extent do you consider that employee awareness promotes process optimisation in Green Hotels?	(Acampora, Lucchetti, et al., 2022; Acampora, Preziosi, et al., 2022)
	AG 4: To what extent do you consider that the involvement of employees in environmental problem solving and the continuous improvement of their awareness influences sustainability?	(Aboramadan & Karatepe, 2021; Oriade et al., 2021)
	AG 5: To what extent do you consider that the environmental concerns of employees influence the formation of positive environmental behavioural intentions?	(Hou & Wu, 2021)
Generative AI (GAI)	GAI 1: Do you consider Generative AI to be able to understand and respond to the individual needs of guests and employees?	(Carvalho & Ivanov, 2023; Demir & Demir, 2023; Dwivedi et al., 2023a; Rathore, 2023)
	GAI 2: Do you consider that chatbots based on Generative AI allow for a more fluid interaction tailored to the individual needs of guests?	(Ülkü, 2023)(An et al., 2024a)
	GAI 3: Do you consider that Generative AI allows the optimisation of internal processes in Green Hotels?	(Çolak, 2023b; Dwivedi et al., 2023b)
	GAI 4: Do you consider Generative AI to significantly improve guest interaction and satisfaction?	(An et al., 2024a; Zhu et al., 2023b) (Demir & Demir, 2023)
Sustainability (SUS)	SUS 1: Do you see green hotels as a driver for greener and more responsible practices?	(Dani et al., 2021; Legrand et al., 2022; Sarma & Baruah, 2024)
	SUS 2: Do you believe that green hotels promote fair and equitable labour practices, which can improve employee morale and well-being?	(Ioannides et al., 2021)
	SUS 3: Do you see green hotels as promoting environmental protection?	(Kholijah, 2024)

Employee Happiness (EHA)	SUS 4: Do you see green hotels as promoting social responsibility?	(Kholijah, 2024)
	SUS 5: To what extent do you consider that green hotels are committed to reducing their environmental and social impact?	(Nisar et al., 2021; Robina-Ramírez, Torrecilla-Pinero, et al., 2023)
	EHA 1: Do you believe that human resource management focused on the development of sustainable and environmentally friendly practices stimulates your job satisfaction?	(Abdelhamied et al., 2023)
	EHA 2: Do you think that the environmentally friendly and sustainable human resource management practices of Green Hotels facilitate employee happiness?	(Choudhary & Datta, 2023)
	EHA 3: Do you consider that the sustainable and green policies of green hotels encourage the subjective and pro-environmental well-being of their employees?	(Gyensare et al., s. f.)
	EHA 4: Do you consider that a pro-environmental organisational climate promotes the subjective well-being of Green Hotel employees?	(Nisar et al., 2024; Patwary et al., 2022)
	EHA 5: Do you believe that environmental commitment and employee confidence in green hotels enhance service excellence?	(Ravina-Ripoll & Robina-Ramírez, 2023)

Table 2. Outer model loadings

	AGH	GAI	EHA	EPG	MPG	SUS
AG1	0,922					
AG5	0,911					
GAI 1		0,898				
GAI 2		0,884				
GAI 3		0,847				
EHA4			0,907			
EHA5			0,933			
EP1				0,914		
EP4				0,887		
EP5				0,877		
MP1					0,927	
MP4					0,940	
SUS1						0,860
SUS3						0,901

SUS4 0,899

Table 3. Cronbach's alpha coefficients, Rho_A, construct reliability and average variance extracted AVE).

	Cronbach's Alpha	rho_A	Composite Reliability	Average Variance Extracted (AVE)
AG	0,810	0,812	0,913	0,840
GAI	0,849	0,849	0,909	0,768
EHA	0,820	0,834	0,917	0,847
EP	0,874	0,887	0,922	0,798
MP	0,853	0,859	0,932	0,872
SUS	0,864	0,866	0,917	0,787

Table 4. Discriminant validity matrix (Fornell-Larcker Criterion)

	AG	GAI	EHA	EP	MP	SUS
AG	0,916					
GAI	0,410	0,877				
EHA	0,456	0,469	0,920			
EP	0,550	0,396	0,682	0,893		
MP	0,609	0,658	0,698	0,609	0,934	
SUS	0,227	0,476	0,550	0,441	0,512	0,887

Table 5. Discriminant validity matrix (Heterotrait-Monotrait Ratio Criterion)

	AG	GAI	EHA	EP	MP	SUS
AG						
GAI	0,494					
EHA	0,561	0,562				
EP	0,655	0,452	0,793			
MP	0,729	0,770	0,832	0,694		
SUS	0,269	0,556	0,652	0,496	0,599	

Table 6. Structural model results

	Rsquare	Sample Mean (M)	Standard Deviation (STDEV)	T Statistics (O/STDEV)	P Values	Q ²
AG	0,168	0,169	0,030	5,642	0,000	0,164
EHA	0,465	0,465	0,049	9,568	0,000	0,186
EP	0,454	0,459	0,048	9,480	0,000	0,152
MP	0,616	0,620	0,036	17,273	0,000	0,430

SUS	0,227	0,230	0,039	5,837	0,000	0,222
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Table 7. Structural model results. Path significance using percentile bootstrap 95% confidence interval (n = 10.000 subsamples)

Hypot thesis	Results	Influence	SPC	Sample Mean (M)	Standard Deviation (STDEV)	T Statistics (O/STDEV)	P Values	Camb io signo
H1	Accepted	GAI -> AG	0,410	0,410	0,036	11,282	0,000	No
H2	Accepted	GAI -> MP	0,380	0,379	0,037	10,249	0,000	No
H3	Accepted	GAI -> SUS	0,476	0,478	0,041	11,657	0,000	No
H4	Accepted	AG -> EP	0,312	0,314	0,052	6,059	0,000	No
H5	Accepted	AG -> MP	0,399	0,398	0,049	8,169	0,000	No
H6	Accepted	SUS -> EP	0,211	0,214	0,051	4,144	0,000	No
H7	Accepted	SUS -> MP	0,241	0,241	0,039	6,143	0,000	No
H8	Accepted	MP -> EP	0,311	0,306	0,064	4,820	0,000	No
H9	Accepted	EP -> EHA	0,682	0,681	0,036	19,029	0,000	No

Table 8. Effects on endogenous variables (extended model)

Dependent Variable	Rsquare	Q ²	Antecedents	PathCoefficients	Correlations	Explained variance(%)
AG	0,168	0,164				16,80
EP	0,454	0,152	H3: GAI	0,410	0,410	16,80
			H1: AG	0,312	0,550	17,16
			H7: MP	0,311	0,609	18,93
			H8: SUS	0,211	0,441	9,30
MP	0,616	0,430				61.60
			H2: AG	0,399	0,609	24,29
			H4: GAI	0,380	0,658	25,00
			H9: SUS	0,241	0,512	12,33

SUS	0,227	0,222				22,70
			H5: GAI	0,476	0,476	22,65
EHA	0,465	0,186				46,50
			H6: EP	0,682	0,682	46,50

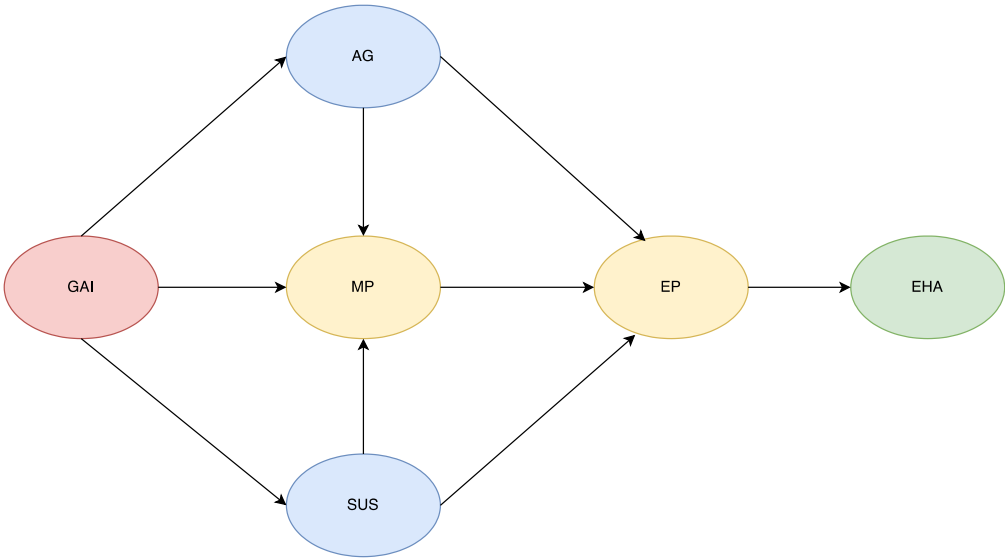


Figure 1. Model constructs

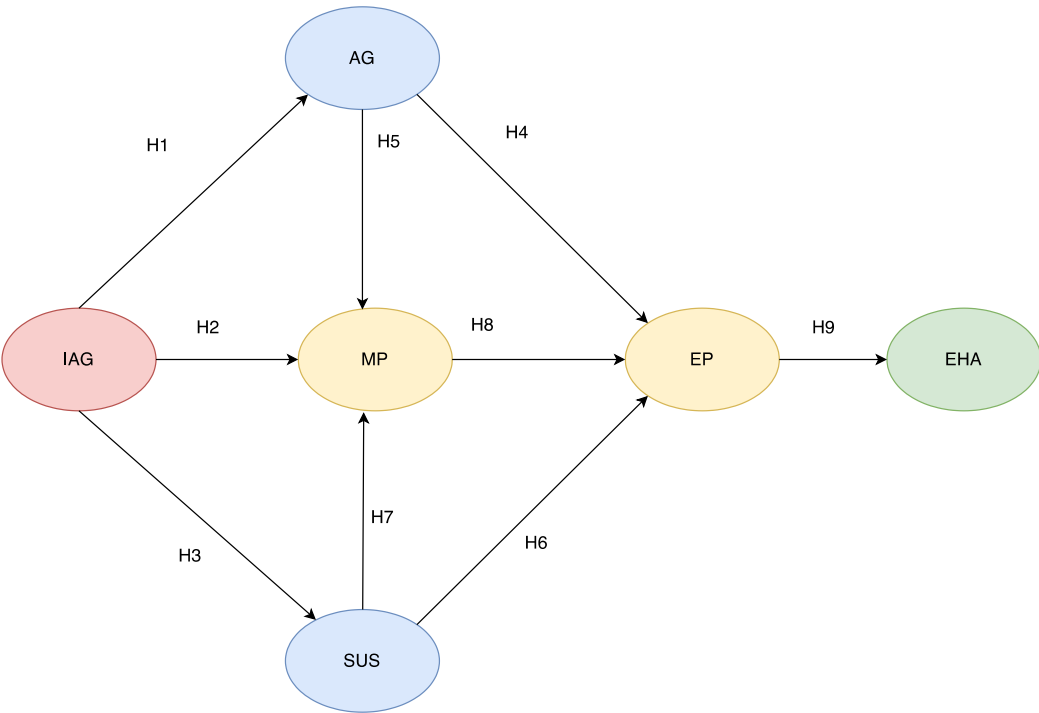


Figure 2. Structural model results (baseline model)

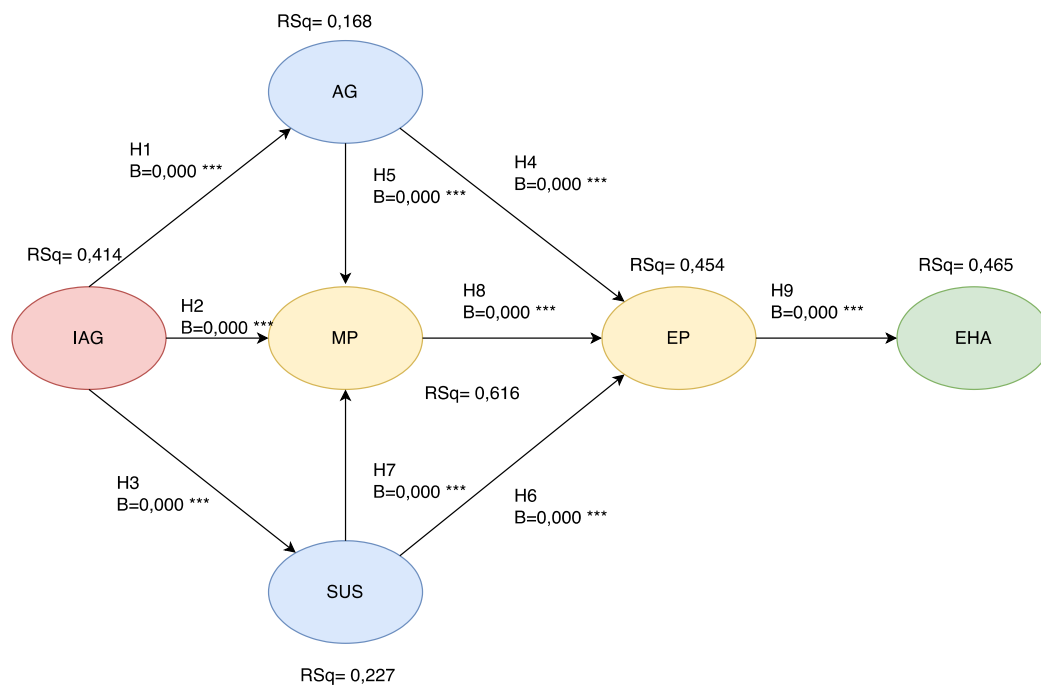


Figure 3: Results of testing the model significance * $p < 0.05$. $P < .01$. * $P < .001$

Associate Editor and Reviewer(s)' Comments to Author:

As requested, we have incorporated four relevant references from your journal, with a particular focus on those published in recent years. These additions have strengthened the foundation of our study and ensured that our work is aligned with the current research trends.

Reviewer: 1

Recommendation: Minor Revision

Thank you very much for taking the time to revise our paper. Your thoughtful feedback and suggestions have been extremely helpful, allowing us to significantly improve the overall quality, clarity, and coherence of the work. We truly appreciate your contribution to enhancing our research and presentation.

Comments: This is a highly relevant and innovative study that addresses a contemporary issue in hospitality management. Your theoretical model is compelling and well-supported by statistical evidence. Strengthen the critical tone in the literature review—move beyond listing to synthesizing themes and gaps. Please revise the manuscript for language and grammar to enhance clarity. Consider briefly discussing the implications for industries beyond tourism to broaden the reach of your findings.

We have strengthened the critical tone of the literature review by moving beyond a simple listing of sources to a more analytical synthesis of key themes and research gaps. Additionally, the jargon and expressions throughout the manuscript have been carefully reviewed and refined by a native English speaker, significantly improving its clarity, fluency, and overall readability. These enhancements contribute to a more polished and professional presentation of our research, making it more accessible to a broader academic audience. Furthermore, we have extended the discussion to include implications for industries beyond tourism, thereby broadening the relevance and potential impact of our findings.

To broaden the applicability of our findings beyond the tourism sector, it is important to recognize that the positive impact of generative artificial intelligence (GAI) on employee happiness and sustainable process management can extend to various industries. Sectors such as healthcare, education, retail, and manufacturing can benefit from GAI by automating routine tasks, promoting environmental awareness, and enhancing job satisfaction through more meaningful work. The integration of GAI can support sustainable human resource practices by fostering a culture of innovation, transparency, and pro-environmental behavior. These improvements not only enhance operational efficiency but also contribute to a more motivated and engaged workforce. Therefore, the insights gained from this study have the potential to inspire cross-sectoral strategies that leverage GAI to align sustainability goals with employee well-being, ultimately contributing to long-term organizational performance and social value creation across industries.

Additional Questions:

1. Originality: Does the paper contain new and significant information adequate to justify publication?: The manuscript presents a timely and original exploration into how generative artificial intelligence (GAI) influences employee happiness and sustainability in green hotel environments within Spain. This focus on the intersection of GAI, sustainability, and happiness—particularly within the tourism and hospitality industry—is novel, especially given the context of Industry 5.0. The study contributes significantly by empirically validating a theoretical model linking GAI with environmental and organizational outcomes in eco-hotels.

Thank you very much.

2. Relationship to Literature: Does the paper demonstrate an adequate understanding of the relevant literature in the field and cite an appropriate range of literature sources? Is any significant work ignored?: The manuscript demonstrates a strong grasp of relevant literature and includes a comprehensive review spanning green hotels, sustainable HRM, AI applications, and happiness management. The citations cover recent contributions and highlight knowledge gaps effectively. However, the manuscript could benefit from slightly more critique and synthesis of literature, rather than descriptive listing.

We sincerely thank the reviewer for acknowledging the breadth and relevance of the literature reviewed in the manuscript, particularly in the areas of green hotels, sustainable HRM, AI applications, and happiness management. In response to the valuable suggestion to move beyond descriptive listing and enhance the level of critique and synthesis, we have revised the literature review section. We believe these changes have strengthened the theoretical foundation of the manuscript and improved the analytical depth of the literature review.

3. Methodology: Is the paper's argument built on an appropriate base of theory, concepts, or other ideas? Has the research or equivalent intellectual work on which the paper is based been well designed? Are the methods employed appropriate?: The use of a mixed-methods approach—focus groups and PLS-SEM—is appropriate and strengthens the robustness of the findings. The model is complex but well-justified, and the data (n=221) appears to be handled rigorously. Measurement validity and reliability are clearly demonstrated through AVE, composite reliability, and discriminant validity checks.

Thank you very much.

4. Results: Are results presented clearly and analysed appropriately? Do the conclusions adequately tie together the other elements of the paper?: The results are clearly presented through tables and diagrams. The hypotheses are tested and discussed with appropriate statistical rigor. The use of R^2 , Q^2 , and path coefficients are clearly explained and support the proposed theoretical relationships.

Thank you very much.

5. Implications for research, practice and/or society: Does the paper identify clearly any implications for research, practice and/or society? Does the paper bridge the gap between theory and practice? How can the research be used in practice (economic and commercial impact), in teaching, to influence public policy, in research (contributing to the body of knowledge)? What is the impact upon society (influencing public attitudes, affecting quality of life)? Are these implications consistent with the findings and conclusions of the paper?: This paper makes strong theoretical and practical contributions. It demonstrates how GAI can support sustainability and employee well-being—both of which are critical in post-COVID and Industry 5.0 contexts. The study effectively bridges theory and practice and includes suggestions for policy, green HRM strategies, and digital transformation in tourism.

Thank you very much.

6. Quality of Communication: Does the paper clearly express its case, measured against the technical language of the field and the expected knowledge of the journal's readership? Has attention been paid to the clarity of expression and readability, such as sentence structure, jargon use, acronyms, etc.: The writing is mostly clear, though at times overly dense and repetitive. The manuscript would benefit from copy-editing to enhance flow and reduce redundancy. Some phrasing appears translated or non-native, which may distract readers.

The jargon and expressions used throughout the paper have been carefully reviewed and corrected by a native English speaker. This has greatly enhanced the clarity, fluency, and overall readability of the manuscript. We believe these improvements contribute to a more professional presentation of our research and make the content more accessible to a wider academic audience.

Reviewer: 2

Recommendation: Major Revision

Thank you very much for taking the time to revise our paper. Your thoughtful feedback and suggestions have been extremely helpful, allowing us to significantly improve the overall quality, clarity, and coherence of the work. We truly appreciate your contribution to enhancing our research and presentation.

Comments:

The paper needs major revisions. The paper examines a relatively new topic. However, the author needs to ensure consistency of terms throughout the paper eg since the paper examines employee happiness, then don't mention on occupational well being and job satisfaction in the abstract, introduction and literature review.

According to your observations I have homogenized the entire article with the construct employee happiness to give it more coherence and solidity based on the conceptual model of the present research. The changes made are underlined in yellow in the paper.

Same applies on pro environmental behaviours - is this construct examined and measured? The author need to include an underlying theory and more in depth

and relevant literature to justify the contributions of the paper. The LR needs to include relevant empirical studies to justify all the hypotheses. The author is requested to check that all citations are relevant for each hypothesis. each construct has to be defined clearly to facilitate understanding.

We sincerely appreciate your insightful comments and the opportunity to clarify aspects of our manuscript and you can find the changes made in blue lettering throughout the manuscript.

Regarding the point about pro-environmental behavior, it is important to clarify that while this concept is central to understanding employee engagement with sustainability, it is not treated as a standalone construct within our model. Instead, the behaviors and attitudes typically associated with pro-environmental conduct are implicitly captured through related constructs such as Awareness, Sustainability Practices, and Management Process, all of which are clearly defined and rigorously measured in the study. These constructs encompass the cognitive and behavioral dimensions of environmental responsibility among employees, as supported by relevant theoretical frameworks such as the Theory of Planned Behavior and Value-Belief-Norm theory. By integrating these perspectives, our model effectively reflects the mechanisms through which employees' environmental attitudes translate into sustainable actions within the hotel context. Furthermore, we have enriched the manuscript's theoretical foundation and literature review to better justify the proposed hypotheses, incorporating recent empirical studies that align directly with each construct and their interrelationships. This ensures that the contributions of the paper are well grounded in both theory and current evidence, addressing the relevance and significance of the constructs in driving green hotel performance and employee outcomes.

We trust that these clarifications and enhancements provide a comprehensive response to your concerns and reinforce the robustness and clarity of the research framework.

Theoretical implications needs significant enhancement - how the study enrich the theory and body of knowledge.

In accordance with your suggestion, we have improved the theoretical implications of the present research in order to enrich the study's theory and body of knowledge.

The changes made are as follows:

Theoretical Implications

This study has important theoretical implications for green human resource management in the green hotel environment. The first is that it is one of the first studies to empirically explore the artificial intelligence-employee happiness binomial in this type of tourist accommodation. In this sense, this academic work contributes to the happiness management literature by showing how the dimensions of artificial intelligence, sustainability and environmental awareness significantly affect employee happiness in the business context of green hotels (Salazar-Altamirano et al., 2025). Second, to demonstrate the theoretical relevance of the environmental performance and management process constructs in the green hotel literature as mediators between the artificial intelligence and employee happiness variables. These variables are key to developing green organisational cultures that generate innovative and sustainable competitive advantages (Martínez-Falcó et al., 2024). Third, to make it clear in tourism

management studies that sustainable practices in green human resource management are very effective in improving the quality of green hotel services when they focus on enriching environmental commitment and employee happiness (Nisar et al., 2024; Robina-Ramírez et al., 2023). Lastly, theoretical research on green hotels should be promoted to address the importance of artificial intelligence and environmental awareness in corporate governance, which stimulates the green intellectual capital of their internal customers (Haldorai et al., 2022).

Additional Questions:

1. Originality: Does the paper contain new and significant information adequate to justify publication?: The paper examines a relatively new topic. However, the author needs to ensure consistency of terms throughout the paper, eg since the paper examines employee happiness, then don't mention on occupational well being and job satisfaction in the abstract, introduction and literature review.

Thank you very much.

According to your suggestion it is mention the relation between **Employee happiness** is directly related to **occupational well-being** and **job satisfaction**. It is also better connected in the section “Employees Hapiness (EHA)” in the literature review how **occupational well-being** and **job satisfaction** affect **employee happiness** by showing how they act as the building blocks of happiness management.

Abstract

Purpose: This study aims to empirically examine the impact of generative artificial intelligence (GAI) on employee happiness within green hotel environments in Spain. By analysing 221 eco-hotels, it explores how GAI influences the awareness of key sustainability indicators, environmental performance, process management, and sustainability—and how these, in turn, affect employee happiness. **Employee happiness is directly related to occupational well-being and job satisfaction, fostering greater motivation, lower turnover, and improved service quality.** The research seeks to fill a gap in the happiness management literature by investigating how advanced technologies can create more sustainable, competitive, and human-centered workplaces in the hospitality sector, especially in the context of Industry 5.0.

That relationship is now more visible in the **literature review**, in the section: **Employee Happiness (EHA)**

Two key elements are necessary for this approach: **First, affective and empathetic communication fosters an inclusive atmosphere where employees feel a sense of belonging, enhancing job satisfaction and supporting occupational well-being.** This inclusive climate encourages proactive participation, strengthens equity, and develops interpersonal relationships, thereby consolidating employee happiness (Farooq et al., 2024; Othman and Elwazer, 2022). **Second, leadership styles focused on emotional intelligence, assertive communication, and disruptive thinking are crucial for building a business philosophy that enriches both professional development and employee happiness.** By addressing well-being and job satisfaction, happy leadership promotes positive emotions, ethical judgment, and sustainable performance, aiming to boost **economic viability** (Ali, 2024; Díaz-García et al., 2024).

Historically, the hospitality industry has been marked by stress and labour tensions, which undermine occupational well-being, reduce job satisfaction, and ultimately diminish employee happiness, negatively affecting service quality and customer experience (Al-Romeedy and Khairy, 2024). **Recent research stresses that hotel**

governance must reverse this trend by nurturing employee happiness as a means of optimizing productivity, commitment, and trust (Bohdanowicz and Zientara, 2009). In this sense, employee happiness—sustained by well-being and job satisfaction—is recognized as an intangible asset that not only improves service quality but also promotes environmental awareness, aligning with Sustainable Development Goals (Arun Kumar et al., 2024).

Same applies on pro environmental behaviours - is this construct examined and measured? The author need to include an underlying theory and more in depth and relevant literature to justify the contributions of the paper.

The paragraph “**Sustainability (SUS)**” include new insights about how sustainability principles directly affect employees’ environmental behaviours in green hotel management:

Sustainability (SUS)

Green hotel management is guided by sustainability principles that minimize environmental and social impacts (Irani et al., 2022; Liu and Hu, 2021; Momayez et al., 2023b). These principles include safer working practices (Irani et al., 2022), waste reduction (Moreno-Luna et al., 2021), efficient use of resources (Dani et al., 2021), and environmentally friendly technologies (Haldorai et al., 2022a; Suárez-Fernández et al., 2024). Recent contributions show that embedding sustainability into operational routines not only enhances environmental performance but also reflects core organisational capabilities that support long-term advantage (Kuo et al., 2022; Mehmood et al., 2023). At the employee level, environmentally responsible behaviour is shaped by internalised values and perceived obligations, as explained by the value-belief-norm theory (Fauzi et al., 2024; Zhang et al., 2025). When employees see sustainability principles consistently applied, they are more likely to adopt pro-environmental habits themselves—such as reducing single-use plastics, monitoring water usage, and ensuring waste segregation—because they internalize these practices as part of their professional and ethical responsibility. In addition to enhancing environmental performance, these initiatives significantly impact employee happiness by fostering a values-driven workplace (Merli et al., 2019; Nisar et al., 2021).

Technologies such as green air and water quality systems enhance both environmental preservation and guest experience, contributing to the hotel’s reputation as a sustainable destination (Jurado-Rivas and Sánchez-Rivero, 2019; Sahoo et al., 2023). Intelligent energy management systems, for instance, optimize energy use and reduce waste, improving operational efficiency (Philipp et al., 2022; Sarma and Baruah, 2024). AI-based environmental monitoring enables proactive resource management, further fostering sustainability (Demir and Demir, 2023; Sahoo et al., 2023). Exposure to these technologies encourages employees to become more environmentally conscious in their daily routines, as they learn to monitor energy consumption, adjust systems for efficiency, and model eco-friendly behaviour to guests. Employees often carry these practices beyond their professional roles, reinforcing a culture of sustainability both inside and outside the workplace.

Sustainability in green hotels promotes a sense of purpose among employees, boosting motivation and employee happiness (Ababneh, 2021; Abuelhassan and Elsayed, 2020). Employees engaged in sustainability initiatives, such as beach clean-ups or reforestation, feel more connected to their work and colleagues, enhancing employee happiness (Henche et al., 2020; Kholijah, 2024; Malik et al., 2021). Participation in these initiatives strengthens pro-environmental behaviour because employees see their actions making a tangible difference. This experiential involvement increases their environmental commitment, reinforcing daily eco-friendly practices such as conserving resources and educating guests on sustainable behaviour. This sense of

community and shared purpose improves co-worker relationships and overall workplace happiness (Al-Sabi et al., 2024; Robina-Ramirez et al., 2023).

Moreover, green hotels often implement fair labour practices, such as inclusive hiring, fair wages, and professional development programs (Ioannides et al., 2021; Patwary et al., 2022). Such practices foster employee morale and a sense of belonging, further increasing employee happiness (Yu et al., 2024; Kıymalıoğlu and Yetkin Özbük, 2023). When employees perceive fairness and respect in management, they are more willing to align with the hotel's sustainability mission, as they see their personal values reflected in organizational practices. This moral alignment leads to stronger engagement in environmental behaviours, from supporting recycling programs to advocating for green policies within the workplace.

2. Relationship to Literature: Does the paper demonstrate an adequate understanding of the relevant literature in the field and cite an appropriate range of literature sources? Is any significant work ignored?: The LR needs to include relevant empirical studies to justify all the hypotheses. The author is requested to check that all citations are relevant for each hypothesis. each construct has to be defined clearly to facilitate understanding.

We thank the reviewer for highlighting the importance of a thorough engagement with relevant **empirical literature** and the clear definition of constructs to support the hypotheses presented. Response is organized through the sections of the LR.

Generative Artificial Intelligence (GAI)

The literature review of the paper supports the hypotheses with relevant empirical evidence. For H1, Carvalho and Ivanov (2023) and Demir and Demir (2023) show that GAI enhances environmental awareness by training staff, monitoring eco-goals, and reminding employees of eco-friendly practices, increasing sustainability consciousness. For H2, Dwivedi et al. (2023a) and Demir and Demir (2023) confirm that GAI optimizes green hotel management processes by automating operations, streamlining workflows, and reducing waste. For H3, Dwivedi et al. (2023a) and Rathore (2023) demonstrate that GAI adoption fosters sustainable practices, strengthening hotels' commitment to eco-responsibility

Based on these arguments, the following research hypothesis is proposed:

H1: Generative Artificial Intelligence (GAI) positively influences Awareness (AG)

H2: Generative Artificial Intelligence (GAI) positively influences Management Process Green Hotels (MP).

H3: Generative Artificial Intelligence (GAI) positively influences Sustainability (SUS).

H6: Sustainability (SUS) positively influences Green Hotels Environmental Performance (EP).

H7: Sustainability (SUS) positively influences Management Process Green Hotels (MP).

Environmental Awareness (AG)

The literature highlights that environmental awareness (AG) in green hotels enhances both environmental performance (EP) and management processes (MP). Han et al. (2020) and Robina-Ramírez et al. (2023) demonstrate that awareness programs foster employees' eco-friendly behaviors, improving resource efficiency and waste reduction,

thus strengthening EP. Similarly, Hameed et al. (2022) and Elbelehy & Crispim (2024) confirm that AG supports structured green practices, aligning staff with sustainability goals and optimizing operational management (MP), reinforcing responsibility and competitiveness in hospitality. Those authors support hypotheses H4 and H5 (Han et al., 2020; Robina-Ramírez et al., 2023; Hameed et al., 2022).

H4: Environmental Awareness (AG) positively influences Green Hotels Environmental Performance (EP).

H5: Environmental Awareness (AG) positively influences Management Process Green Hotels (MP).

Sustainability (SUS)

Empirical studies confirm that sustainability (SUS) strengthens both environmental performance (EP) and management processes (MP) in green hotels. Irani et al. (2022) and Liu & Hu (2021) show that eco-friendly routines, such as energy and water conservation, directly enhance EP by reducing ecological impact. Similarly, Henche et al. (2020) and Nisar et al. (2021) demonstrate that embedding sustainable practices into daily operations improves MP, fostering efficiency, employee engagement, and competitiveness. These findings justify H6 and H7

H6: Sustainability (SUS) positively influences Green Hotels Environmental Performance (EP).

H7: Sustainability (SUS) positively influences Management Process Green Hotels (MP).

Environmental Performance (EP)

Research shows that management processes (MP) strongly improve environmental performance (EP) in green hotels. Ahmed et al. (2021) and Merli et al. (2019) highlight that energy conservation and waste reduction strengthen ecological outcomes. Similarly, Nisar et al. (2022) and Momayez et al. (2023b) confirm MP enhances efficiency and sustainability. Those authors support hypothesis H8.

H8: Management Process Green Hotels (MP) positively influences Green Hotels Environmental Performance (EP).

Environmental Performance (EP)

Research shows that management processes (MP) strongly improve environmental performance (EP) in green hotels. Ahmed et al. (2021) and Merli et al. (2019) highlight that energy conservation and waste reduction strengthen ecological outcomes. Similarly, Nisar et al. (2022) and Momayez et al. (2023b) confirm MP enhances efficiency and sustainability.

H9: Employee happiness (EP) in green hotels is positively influenced by Environmental performance (EHA)

Studies confirm that environmental performance (EP) fosters employee happiness in green hotels. Abuelhassan and Elsayed (2020) show that eco-friendly practices enhance job satisfaction, while Arun Kumar et al. (2024) highlight that pro-environmental behaviors improve well-being. High EP provides purpose and pride, reinforcing happiness through sustainability. Based on these theoretical arguments, the following research hypothesis is formulated:

H9: Employee happiness (EP) in green hotels is positively influenced by Environmental performance (EHA)

3. Methodology: Is the paper's argument built on an appropriate base of theory, concepts, or other ideas? Has the research or equivalent intellectual work on which the paper is based been well designed? Are the methods employed appropriate?: The data analysis method is appropriate. This section needs to include more details such as sampling method, data collection procedures, measurement of constructs. Please make sure that the constructs are measured using validated measurements.

We thank the reviewer for emphasizing the importance of using or adapting all measurement items from previously validated scales.

In this study, all measurement items were adapted from previously validated scales published in peer-reviewed academic journals in the fields of hospitality, sustainability, and organizational behavior. The adaptation process involved rephrasing the items to reflect hotel-specific terminology, operational routines, and employee experiences in eco-certified establishments. Additionally, the language was simplified to ensure clarity and accessibility for employees with diverse educational backgrounds.

The primary goal of our research was to conceptualize and empirically validate the profile of the Employee Happiness (EHA) in Spain. To ensure both theoretical rigor and practical relevance, the study followed a sequential, three-step process aligned with the Resource-Based View (RBV), while grounding findings in the lived experiences of hotel managers.

Step 1: Qualitative Exploration:

The qualitative phase of the study began by presenting ten constructs drawn from the literature review to representatives of twelve randomly selected green hotels. These constructs were Generative Artificial Intelligence (GAI), Environmental Awareness (AG), Sustainability (SUS), Management Processes (MP), Environmental Performance (EP), Work-Life Balance (WLB), Organizational Support (OS), Green Organizational Culture (GOC), Job Autonomy (JA), and Social Cohesion and Teamwork (SCT). The first focus group aimed to define these constructs in the context of green hotel operations, ensuring that each dimension was relevant to both employee experiences and organizational practices. Collectively, they cover technological, environmental, and human-centered aspects that influence Employee Happiness (EHA), providing a comprehensive framework for understanding well-being and sustainable performance in eco-friendly hospitality environments.

Following a deep review during the second focus group, five of the ten constructs—WLB, OS, GOC, JA, and SCT—were removed due to conceptual overlap with the remaining core constructs. WLB was largely captured by indicators of employee well-being within MP and EHA measures. OS was partially integrated into the MP and EHA constructs, reflecting support and recognition without requiring a separate dimension. GOC overlapped with EP and AG in reflecting alignment of employee values with sustainability and environmental practices. JA was conceptually embedded within GAI and MP indicators, which addressed employee empowerment and decision-making in green processes. Finally, SCT was captured through AG and MP constructs in relation to collaboration, engagement, and participation in sustainability initiatives. By removing these five constructs, the model was streamlined without losing theoretical richness or practical relevance, leaving GAI, AG, SUS, MP, and EP as the core constructs directly defining EHA in green hotels.

Step 2: Indicator Development:

Initially, a list of 29 potential indicators was identified from the literature review and prior qualitative insights, designed to capture observable behaviors and perceptions across the five core constructs—Generative Artificial Intelligence (GAI), Environmental Awareness (AG), Sustainability (SUS), Management Processes (MP), and Environmental Performance (EP)—all of which define Employee Happiness (EHA) in green hotels. Each construct included multiple items to comprehensively operationalize the theoretical dimensions: MP1–MP5 evaluated management effectiveness, sustainable materials, leadership, and community collaboration; EP1–EP6 assessed responsibility, culture, policy implementation, alignment with environmental principles, and employee participation; AG1–AG7 captured global awareness, training, process optimization, problem-solving, and behavioral intentions; GAI1–GAI6 measured personalization, chatbot interactions, process optimization, and guest satisfaction; SUS1–SUS5 reflected responsible practices, fair labor, environmental protection, and social responsibility. EHA1–EHA5 addressed the role of sustainable human resource management in fostering employee happiness.

After careful review during the second focus group, indicators exhibiting conceptual overlap or redundancy were removed to streamline the model while preserving conceptual richness. Specifically, MP5, which assessed the contribution of sustainability actions to long-term environmental goals, was removed due to overlap with MP1 on sustainability practices' effectiveness. EP6, concerning the influence of organizational culture on daily environmentally responsible behaviors, was eliminated because it duplicated EP2 and EP5. AG6, regarding continuous awareness campaigns, and AG7, measuring staff ecological awareness, were removed for redundancy with AG2, AG4, and AG5. Similarly, GAI5, which addressed process automation, and GAI6, focused on personalized digital solutions, overlapped with GAI3, GAI1, and GAI4 and were therefore excluded. Lastly, SUS5, evaluating commitment to reducing environmental and social impact, was removed because its content was already captured by SUS1, SUS3, and SUS4.

For instance, MP5 *To what extent do you consider that the sustainability actions of Green Hotels contribute effectively to long-term environmental goals?* (removed due to overlap with MP1 on sustainability practices' effectiveness). EP6 (removed): *To what extent does environmental responsibility embedded in organizational culture influence daily work behaviors?* (removed as it repeated EP2 and EP5 on culture's influence). AG6 (removed): *To what extent do continuous awareness campaigns reinforce employees' ecological commitment?* (removed for redundancy with AG2 and AG4). AG7 (removed): *To what extent does staff ecological awareness directly influence their pro-environmental workplace behaviors?* (removed as it overlapped with AG5). GAI5 (removed): *Do you consider Generative AI to be an efficient tool for process automation and task reduction in Green Hotels?* (removed due to similarity with GAI3). GAI6 (removed): *Do you consider Generative AI to provide personalized digital solutions that enhance the overall guest experience?* (removed due to overlap with GAI1 and GAI4). SUS5 (removed): *To what extent do you consider that green hotels are committed to reducing their environmental and social impact?* (removed because its focus was already captured in SUS1, SUS3, and SUS4).

This careful refinement reduced the original pool from 29 to 22 indicators, ensuring that each item contributed uniquely to operationalizing the constructs without inflating the model or introducing multicollinearity. The resulting indicators provide a precise, measurable foundation for analyzing how GAI, environmental awareness, sustainability, management processes, and environmental performance collectively influence

employee happiness, supporting robust quantitative analysis while remaining closely grounded in qualitative insights.

Step 3: Managerial Interviews: The third step involved semi-structured interviews with employees to validate and contextualize the refined constructs and indicators. These interviews explored how **EHA** manifested in daily operations, providing concrete examples and practical nuances that purely quantitative measures might overlook.

These interviews confirmed that the 22 indicators were both observable and measurable, and that the four constructs— GAI, AG, SUS, MP, EP. —effectively captured the multidimensional nature of EHA. The qualitative insights provided a practical grounding, ensuring that the conceptual model was not only theoretically robust but also applicable across diverse hotel contexts (see Graph 1 and Figure 1).

4. Results: Are results presented clearly and analysed appropriately? Do the conclusions adequately tie together the other elements of the paper?: Results are presented well

Thank you very much.

5. Implications for research, practice and/or society: Does the paper identify clearly any implications for research, practice and/or society? Does the paper bridge the gap between theory and practice? How can the research be used in practice (economic and commercial impact), in teaching, to influence public policy, in research (contributing to the body of knowledge)? What is the impact upon society (influencing public attitudes, affecting quality of life)? Are these implications consistent with the findings and conclusions of the paper?: Theoretical implications needs significant enhancement - how the study enrich the theory and body of knowledge.

In accordance with your suggestion, we have improved the theoretical implications of the present research in order to enrich the study's theory and body of knowledge.

The changes made are as follows:

Theoretical Implications

This study has important theoretical implications for green human resource management in the green hotel environment. The first is that it is one of the first studies to empirically explore the artificial intelligence-employee happiness binomial in this type of tourist accommodation. In this sense, this academic work contributes to the happiness management literature by showing how the dimensions of artificial intelligence, sustainability and environmental awareness significantly affect employee happiness in the business context of green hotels (Salazar-Altamirano et al., 2025). Second, to demonstrate the theoretical relevance of the environmental performance and management process constructs in the green hotel literature as mediators between the artificial intelligence and employee happiness variables. These variables are key to developing green organisational cultures that generate innovative and sustainable competitive advantages (Martínez-Falcó et al., 2024). Third, to make it clear in tourism management studies that sustainable practices in green human resource management are very effective in improving the quality of green hotel services when they focus on enriching environmental commitment and employee happiness (Nisar et al., 2024; Robina-Ramírez et al., 2023). Lastly, theoretical research on green hotels should be

promoted to address the importance of artificial intelligence and environmental awareness in corporate governance, which stimulates the green intellectual capital of their internal customers (Haldorai et al., 2022).

6. Quality of Communication: Does the paper clearly express its case, measured against the technical language of the field and the expected knowledge of the journal's readership? Has attention been paid to the clarity of expression and readability, such as sentence structure, jargon use, acronyms, etc.: Acceptable.

Thank you very much

Generative Artificial Intelligence as a Catalyst for Employee Happiness and Sustainable Performance in Green Hotel Management

Abstract

Purpose: This study examines the impact of generative artificial intelligence (GAI) on employee happiness in 221 Spanish green hotels. It explores how GAI enhances sustainability awareness, environmental performance, and process management, shaping occupational well-being and job satisfaction. *Employee happiness drives motivation, lowers turnover, and improves service quality.* The research shows how advanced technologies foster sustainable, competitive, and human-centered workplaces in hospitality while addressing key gaps in happiness management research.

Design/methodology/approach: This study uses a mixed-method approach combining focus group insights and PLS-SEM analysis of 221 Spanish Ecocertified green hotels, examining how generative AI impacts employee happiness, environmental performance, management processes, and sustainability, providing a novel framework for Industry 5.0 competitiveness.

Findings: The study confirms that generative AI positively influences employee happiness in green hotels by enhancing process efficiency, sustainability, and environmental awareness. GAI improves management processes (61.6% variance), supports sustainable practices, and enables fulfilling work. Environmentally driven performance significantly increases happiness, validating GAI and sustainability as key predictors for a competitive, responsible tourism sector.

Originality: This first EU study links generative AI to employee happiness in Spain’s green hospitality sector, validating a PLS-SEM model that shows GAI fosters pro-environmental behavior, operational efficiency, and well-being, challenging the view of AI as purely job-automation.

Keywords: Green hotels, happiness, sustainability, generative artificial intelligence, technology

Introduction

Corporate governance in hospitality increasingly recognizes that employee happiness drives long-term economic benefits (Ravina-Ripoll et al., 2024; Westover et al., 2010). In green hotels, happiness enhances technological adoption, environmental awareness, and sustainable practices, generating innovation and long-

term value (Abbasi et al., 2024; Bohdanowicz & Zientara, 2009). The Resource-Based View (RBV) highlights how intangible assets—intellectual capital, technological capabilities, and organizational competencies—become competitive advantages when valuable, rare, inimitable, and non-substitutable (Ofori & Appiah-Nimo, 2022; Mehmood et al., 2023). Applied to green hotels, these resources integrate generative AI with environmental strategies, enhancing efficiency and well-being (Kuo, Fang, & LePage, 2022). Employee happiness thus emerges as both an outcome and a resource, reinforcing innovation, resilience, and competitiveness.

Generative artificial intelligence (GAI) represents a transformative RBV capability. In green hotels, it automates tasks, optimizes workflows, and supports real-time decisions, improving efficiency and reducing stress (An et al., 2024a; Demir & Demir, 2023). GAI also personalizes guest experiences and promotes sustainability, enabling staff to embed eco-friendly practices into routines. This strengthens human capital and workplace happiness, though concerns persist about AI replacing human labor (Folgado-Fernández et al., 2025; Ooi et al., 2025).

Sustainability practices further boost happiness. Initiatives such as energy and water conservation, waste reduction, and eco-materials improve environmental performance while enhancing purpose and engagement (Irani et al., 2022; Liu & Hu, 2021). Participation in reforestation or recycling programs builds morale and social contribution (Henché et al., 2020; Malik et al., 2021). Green HRM practices—training, fair labor policies, and participatory decision-making—strengthen commitment and satisfaction (Pham et al., 2018; Aboramadan & Karatepe, 2021). Embedding sustainability into organizational culture aligns personal and professional values, reinforcing competitiveness.

Integrating RBV with behavioral theories, such as planned behaviour and value-belief-norm, clarifies how environmental attitudes shape happiness and sustainability engagement (Fauzi et al., 2024; Wang et al., 2023). Employees' awareness, combined with technological and organizational capabilities, creates a cycle where sustainable practices, performance, and satisfaction reinforce each other.

This study investigates Spanish eco-hotels, key to post-COVID-19 recovery (Leal-Solís & Robina-Ramírez, 2022). Findings show that strategically deployed

intangible resources, strengthened by GAI, enhance employee well-being, competitiveness, and environmental performance. Ultimately, the research contributes to happiness management, sustainable innovation, and AI adoption, demonstrating that GAI and sustainability align with SDGs by positioning happiness as both driver and outcome of strategic management.

Literature Review

This study’s conceptual model examines how generative AI influences employee happiness in green hotels, addressing a gap in emerging literature on Industry 5.0 and employee well-being in sustainable hospitality.

Figure 1. Model constructs

Generative Artificial Intelligence (GAI)

Within the Resource-Based View (RBV), Generative Artificial Intelligence (GAI) is a strategic capability that transforms intangible resources into competitive advantage. By personalizing guest experiences, offering predictive insights, and automating tasks, GAI improves workflow efficiency and reduces employee stress (Ülkü, 2023; Demir & Demir, 2023; Zhu et al., 2023). It also supports sustainability by educating staff and guests, monitoring eco-friendly practices, and enabling continuous environmental improvements (Chung, 2020; Elias et al., 2024). This synergy of technology and organizational knowledge enhances engagement, operational performance, and environmental outcomes, generating multidimensional competitive advantage for green hotels (Wang, 2022; Young & Shishido, 2023).

As a branch of AI that produces contextually relevant outputs, GAI improves organizational and customer interactions (An et al., 2024a). In green hotels, chatbots and virtual assistants personalize experiences, promote sustainable activities, and communicate eco-initiatives (Demir & Demir, 2023; Carvalho & Ivanov, 2023; Rathore, 2023). Beyond guest satisfaction, GAI strengthens internal capabilities by streamlining communication, automating repetitive tasks, and supporting real-time decisions (Workman et al., 2023). This shift allows employees to focus on high-value activities, boosting happiness, reducing stress, and reinforcing human capital as a vital intangible asset.

GAI also facilitates sustainability integration by educating guests, encouraging participation in recycling and energy-saving programs, and gathering feedback to refine eco-strategies (Chung, 2020; Martín Lucas et al., 2025; Paul et al., 2023). Combining technological capabilities with organizational knowledge enhances environmental performance and aligns both employee and guest engagement with sustainability goals (Zhu et al., 2023; An et al., 2024b).

Strategic GAI deployment exemplifies RBV by converting advanced technology into competitive advantage, improving efficiency, environmental performance, and employee happiness (Wang, 2022; Young & Shishido, 2023). As both a technological and human-capital enhancer, GAI enables green hotels to translate intangible and tangible assets into measurable value, fostering a cycle where innovation, sustainability, and workforce satisfaction reinforce one another.

The literature supports these links: for H1, GAI raises environmental awareness by training staff and monitoring eco-goals (Carvalho & Ivanov, 2023; Demir & Demir, 2023). For H2, it optimizes management through automation and waste reduction (Dwivedi et al., 2023a; Demir & Demir, 2023). For H3, GAI fosters sustainable practices, reinforcing hotels' eco-responsibility (Dwivedi et al., 2023a; Rathore, 2023).

Based on these arguments, the following research hypothesis is proposed:

H1: Generative Artificial Intelligence (GAI) positively influences Awareness (AG)

H2: Generative Artificial Intelligence (GAI) positively influences Management Process Green Hotels (MP).

H3: Generative Artificial Intelligence (GAI) positively influences Sustainability (SUS).

Environmental Awareness (AG)

The hospitality sector consumes high levels of energy and water and generates significant waste, posing challenges for sustainable operations (Han et al., 2020; Robina-Ramírez et al., 2021; Luo et al., 2021). From a Resource-Based View (RBV) perspective, environmental awareness constitutes a strategic intangible resource that strengthens human capital, operational efficiency, and competitive advantage while enhancing employee well-being (Ofori and Appiah-Nimo, 2022; Mehmood et al., 2023; Kuo, Fang,

and LePage, 2022). In green hotels, employees with strong environmental awareness integrate sustainable practices into daily operations, improving processes, reducing waste, and fostering a culture of responsibility.

Green hotels also cultivate environmental consciousness among guests, prompting them to adopt eco-friendly behaviors that reinforce the hotel’s green image and operational sustainability (Cembruch-Nowakowski, 2019; Acampora, Preziosi, et al., 2022). Structured awareness programs educate both employees and customers on best practices, aligning personal and professional objectives with sustainability goals and contributing to long-term competitive advantage (Han et al., 2020; Elbelehy and Crispim, 2024; Robina-Ramírez et al., 2023).

Human resource strategies within green hotels leverage RBV by embedding sustainability-focused training, incentives, and participatory decision-making processes, strengthening employee commitment to environmental initiatives and workplace happiness (Pham et al., 2018; Alreahi et al., 2023; M. Ahmed et al., 2020; Aboramadan and Karatepe, 2021). Generative AI (GAI) supports this resource-based approach by providing real-time updates, interactive challenges, and dashboards that facilitate eco-friendly practices, enhance engagement, and reinforce organizational culture (Demir and Demir, 2023; Hou and Wu, 2021).

The literature highlights that environmental awareness (AG) in green hotels enhances both environmental performance (EP) and management processes (MP). Han et al. (2020) and Robina-Ramírez et al. (2023) demonstrate that awareness programs foster employees’ eco-friendly behaviors, improving resource efficiency and waste reduction, thus strengthening EP. Similarly, Hameed et al. (2022) and Elbelehy & Crispim (2024) confirm that AG supports structured green practices, aligning staff with sustainability goals and optimizing operational management (MP), reinforcing responsibility and competitiveness in hospitality. Those authors support hypotheses H4 and H5 (Han et al., 2020; Robina-Ramírez et al., 2023; Hameed et al., 2022).

H4: Environmental Awareness (AG) positively influences Green Hotels Environmental Performance (EP).

H5: Environmental Awareness (AG) positively influences Management Process Green Hotels (MP).

Sustainability (SUS)

Green hotel management follows sustainability principles aimed at minimizing environmental and social impacts (Irani et al., 2022; Liu and Hu, 2021). Practices include safer working conditions (Irani et al., 2022), waste reduction (Moreno-Luna et al., 2021), efficient resource use (Dani et al., 2021), and eco-friendly technologies (Haldorai et al., 2022a; Suárez-Fernández et al., 2024). From a Resource-Based View (RBV), embedding sustainability into operations strengthens organizational capabilities, providing long-term competitive advantages (Kuo et al., 2022; Mehmood et al., 2023). Employees internalize these practices, adopting pro-environmental habits such as reducing single-use plastics, monitoring water, and segregating waste, which also enhances workplace happiness (Merli et al., 2019; Nisar et al., 2021).

Technologies like green air and water quality systems, intelligent energy management, and AI-based environmental monitoring optimize resource use and improve operational efficiency (Jurado-Rivas and Sánchez-Rivero, 2019; Philipp et al., 2022; Sarma and Baruah, 2024; Demir and Demir, 2023; Sahoo et al., 2023). These systems encourage employees to adopt eco-friendly behaviours in their routines and model sustainability for guests, reinforcing organizational environmental culture.

Engagement in sustainability initiatives, such as reforestation or beach clean-ups, fosters purpose, motivation, and workplace happiness among staff (Ababneh, 2021; Abuelhassan and Elsayed, 2020; Henche et al., 2020; Kholijah, 2024; Malik et al., 2021; Al-Sabi et al., 2024; Robina-Ramirez et al., 2023). Experiential participation strengthens pro-environmental behaviour, improves co-worker relationships, and promotes a sense of community.

Inclusive HR practices, including fair wages, professional development, and equitable treatment, further enhance employee morale and alignment with sustainability goals (Ioannides et al., 2021; Patwary et al., 2022; Yu et al., 2024; Kıymalıoğlu and Yetkin Özbük, 2023). Employees who perceive fairness are more engaged in green initiatives, supporting recycling programs and advocating sustainability internally.

Empirical studies confirm that sustainability (SUS) strengthens both environmental performance (EP) and management processes (MP) in green hotels. Irani et al. (2022) and Liu & Hu (2021) show that eco-friendly routines, such as energy and water conservation, directly enhance EP by reducing ecological impact. Similarly, Henche et al. (2020) and Nisar et al. (2021) demonstrate that embedding sustainable practices into daily operations improves MP, fostering efficiency, employee engagement, and competitiveness. These findings justify H6 and H7

H6: Sustainability (SUS) positively influences Green Hotels Environmental Performance (EP).

H7: Sustainability (SUS) positively influences Management Process Green Hotels (MP).

Management Process (MP)

Green hotels provide an innovative response to hospitality’s environmental challenges (Ahmed et al., 2021). They implement energy conservation, waste reduction, water efficiency, and sustainable material use, often collaborating with local communities to promote sustainability (Merli et al., 2019; Nisar et al., 2021, 2022; Acampora, Lucchetti, et al., 2022; Koval et al., 2023). From a Resource-Based View, these practices reflect core organizational capabilities, turning sustainability into a strategic strength that supports long-term competitive advantage (Mehmood et al., 2023; Ofori and Appiah-Nimo, 2022).

Generative Artificial Intelligence (GAI) enhances sustainability management in green hotels by analyzing guest data, optimizing energy use, and automating inventory to reduce waste (An et al., 2024a; Demir and Demir, 2023; Ülkü, 2023; Zhu et al., 2023). AI-driven learning platforms provide interactive sustainability training, fostering employee knowledge, engagement, and commitment to eco-friendly practices (Elias et al., 2024; Demir and Demir, 2023). Sustainability-focused HR policies further reinforce pro-environmental behavior while enhancing employee happiness by creating safe, purpose-driven workplaces (Haldorai et al., 2022a; Yong et al., 2022; Nisar et al., 2021, 2022; Ravina-Ripoll and Robina-Ramírez, 2023).

AI integration enables real-time monitoring of KPIs related to energy efficiency, waste, and carbon footprint, promoting continuous improvement and operational success (Paul et al., 2023). By linking sustainability, employee engagement, and technological capability, green hotels foster talent retention, productivity, and a virtuous cycle where environmental performance and employee happiness mutually reinforce one another (Suárez-Fernández et al., 2024). Research shows that management processes (MP) strongly improve environmental performance (EP) in green hotels. Ahmed et al. (2021) and Merli et al. (2019) highlight that energy conservation and waste reduction strengthen ecological outcomes. Similarly, Nisar et al. (2022) confirm MP enhances efficiency and sustainability. Those authors support hypothesis H8.

H8: Management Process Green Hotels (MP) positively influences Green Hotels Environmental Performance (EP).

Environmental Performance (EP)

Environmental performance is a key indicator of green hotels' commitment to sustainability (Nisar et al., 2021). From a Resource-Based View (RBV), it results from the strategic deployment of intangible resources—green knowledge, technological capabilities, and organizational competencies—that create sustained competitive advantage (Ofori and Appiah-Nimo, 2022; Mehmood et al., 2023). Employees' pro-environmental behaviour, shaped by internalized values and norms, contributes directly to environmental performance and links it to workplace satisfaction and happiness (Abuelhassan and Elsayed, 2020; Arun Kumar et al., 2024). High environmental performance demonstrates successful implementation of eco-friendly strategies, improving ecological impact and market competitiveness as sustainability-conscious travelers increasingly prefer environmentally responsible accommodations (Carvalho and Ivanov, 2023; Raza and Khan, 2022).

Innovative technologies, particularly generative artificial intelligence (GAI), enhance environmental performance by enabling real-time energy monitoring, predictive maintenance, and automated waste tracking (Zhu et al., 2023b). These tools allow data-driven decisions that optimize sustainability and operational efficiency. Employees engaged in eco-friendly initiatives—reducing resource waste, implementing recycling programs, and guiding guest participation—experience greater job satisfaction, as they integrate sustainability into daily work tasks (Chung, 2020; Nisar et al., 2021, 2022). AI-enhanced adaptive training further strengthens environmentally responsible behaviours, improving employees' skills and engagement (Elias et al., 2024).

Green intellectual capital—employees' knowledge and competencies regarding sustainable practices—is central to sustaining environmental improvements (Haldorai et al., 2022a; Wei et al., 2023). Investing in this capital through training and fostering a sustainability-oriented organizational culture ensures a workforce capable of driving long-term eco-performance while increasing employee happiness and sense of purpose (Abuelhassan and Elsayed, 2020; Ravina-Ripoll et al., 2024; Arun Kumar et al., 2024).

Employee Happiness (EHA)

In recent decades, research on organizational happiness has expanded, particularly through the philosophy of happiness management (Kamel et al., 2017; Rando Cueto et al., 2023). Workplace happiness—rooted in occupational well-being and job satisfaction—is recognized as an intangible resource that boosts productivity while

generating competitive and ecological advantages (Kıymalıoğlu & Yetkin Özbük, 2023). Effective management requires strategic direction that prioritizes sustainability and human capital well-being (Malik et al., 2021), enabling organizations to adopt models that promote corporate happiness (Robina-Ramirez et al., 2023).

Two factors are central to this approach. First, affective and empathetic communication fosters inclusivity, belonging, and proactive participation, strengthening equity and interpersonal relationships while consolidating employee happiness (Farooq et al., 2024; Othman & Elwazer, 2022). Second, leadership styles based on emotional intelligence, assertive communication, and disruptive thinking enrich professional development and well-being. Happy leadership encourages positive emotions, ethical judgment, and sustainable performance, thereby strengthening economic viability (Ali, 2024; Díaz-García et al., 2024).

The hospitality sector has historically been characterized by stress and labor tensions, reducing well-being, job satisfaction, and employee happiness, which harms service quality and customer experience (Al-Romeedy & Khairy, 2024). Recent studies emphasize that hotel governance must reverse this trend by fostering happiness to optimize productivity, commitment, and trust (Bohdanowicz & Zientara, 2009). As an intangible asset, employee happiness enhances service quality, promotes environmental awareness, and aligns with the Sustainable Development Goals (Arun Kumar et al., 2024). In green hotels, happiness reinforces service excellence and workplace climate (Choudhary & Datta, 2023). Management should therefore adopt green policies that simultaneously improve ecological awareness, well-being, and job satisfaction, strengthening employee happiness (Gyensare et al.; Yu et al., 2024).

Studies confirm that environmental performance fosters employee happiness in green hotels. Eco-friendly practices enhance satisfaction (Abuelhassan & Elsayed, 2020), while pro-environmental behaviors improve well-being (Arun Kumar et al., 2024). High environmental performance provides purpose and pride, reinforcing happiness through sustainability.

Based on these theoretical arguments, the following research hypothesis is formulated:

H9: Employee happiness (EP) in green hotels is positively influenced by Environmental performance (EHA)

Methodology

A mixed-methods approach was used, combining qualitative and quantitative techniques to examine employee happiness (EHA) in Spanish eco-hotels. The study focused on Ecostars-certified hotels, which comply with sustainability standards endorsed by the GSTC and UNWTO. Its main objective was to conceptualize and empirically validate the profile of Employee Happiness in Spain. To ensure theoretical rigor and practical relevance, the study followed a three-step sequential process framed by the Resource-Based View (RBV) and informed by hotel managers' lived experiences.

Step 1: Qualitative Exploration

The study began by presenting ten constructs drawn from the literature—Generative Artificial Intelligence (GAI), Environmental Awareness (AG), Sustainability (SUS), Management Processes (MP), Environmental Performance (EP), Work-Life Balance (WLB), Organizational Support (OS), Green Organizational Culture (GOC), Job Autonomy (JA), and Social Cohesion and Teamwork (SCT)—to twelve randomly selected green hotels. Focus groups defined these constructs in the context of green hotel operations. They covered technological, environmental, and human-centered aspects influencing EHA, providing a comprehensive framework for employee well-being and sustainable performance.

During a second focus group, five constructs (WLB, OS, GOC, JA, SCT) were removed due to overlap. WLB was captured within MP and EHA; OS aligned with MP and EHA; GOC overlapped with EP and AG; JA was reflected in GAI and MP; SCT was covered through AG and MP. This refinement left five core constructs—GAI, AG, SUS, MP, and EP—directly defining EHA in green hotels.

Step 2: Indicator Development

An initial pool of 29 indicators, identified through literature and qualitative insights, captured observable behaviors across the five constructs. For example, MP1–MP5 assessed management and leadership, EP1–EP6 examined policy implementation and participation, AG1–AG7 evaluated training and behavioral intentions, GAI1–GAI6 measured personalization and optimization, and SUS1–SUS5 addressed responsible practices. EHA1–EHA5 reflected the role of sustainable HRM.

After review, redundant indicators were removed, reducing the pool to 22. Examples included dropping MP5, EP6, AG6–AG7, GAI5–GAI6, and SUS5, each overlapping with stronger indicators in their construct. The streamlined set ensured conceptual clarity and minimized multicollinearity while retaining theoretical richness. The final indicators

offered a precise foundation for quantitatively analyzing how GAI, AG, SUS, MP, and EP collectively shape employee happiness.

Step 3: Managerial Interviews

Semi-structured interviews with hotel employees validated the refined constructs and indicators. These interviews illustrated how EHA was expressed in daily operations, providing concrete examples that quantitative data alone could not capture. Respondents confirmed that the 22 indicators were both observable and measurable, and that the five constructs—GAI, AG, SUS, MP, and EP—effectively represented the multidimensional nature of EHA.

The qualitative insights grounded the model in practical reality, ensuring it was theoretically robust yet adaptable to diverse hotel contexts. Together, the three phases established a validated framework for analyzing how green hotels integrate advanced technologies, sustainability practices, and management processes to foster employee happiness while enhancing competitiveness and environmental performance (see Table 1).

Table I. Items

Sample and population

Out of the 1,200 hotels contacted via email and telephone, 221 valid responses were collected, yielding a response rate of 18.4%. The response pool adequately represents a range of hotel categories, geographic locations, and sizes within the certified population.

Data processing

This study used latent variable regression via PLS (SmartPLS 4.1.0.3) to estimate structural model parameters, minimizing residual variance (Hair et al., 2013). PLS, suitable for small samples and non-parametric conditions (Hulland, 1999), assessed direct, indirect, and moderating effects of GAI on employee happiness and sustainability.

Results

Hair *et al.* (2014) recommend that the kurtosis and skewness values of the indicators should be within the acceptable range of -1 to +1. Although several values show a slight degree of non-normality, this is not considered problematic, as the degree of skewness is

not severe. Moreover, since there are two indicators measuring the (reflective) construct, this slight deviation from normality does not compromise the validity of the model.

The external loadings indicate the correlations between latent and observed variables, so the values should be higher than 0.7 (Carmines and Zeller, 1979), as can be seen in Table II.

Table II. Outer model loadings

The simple reliability of the measurement scales used was assessed by Cronbach's alpha values, which exceeded 0.70 in all cases (Nunnally and Bernstein, 1994). As for the composite reliability, all indicator values were greater than 0.7 (Werts *et al.*, 1974), indicating a high level of internal consistency among the latent variables.

In the analysis of variance, all mean variance extracted (AVE) values were greater than 0.50 (Fornell and Larcker, 1981), exceeding the minimum acceptable values for validity (Table III).

Table III. Cronbach's alpha coefficients, Rho_A, construct reliability and average variance extracted AVE).

Additionally, Fornell and Larcker (1981) suggest that the square root of the AVE on each latent variable can be used to establish discriminant validity. To confirm discriminant validity between constructs, the square root of the AVE must be higher than the correlation between the constructs. Table IV presents the square roots of the AVE on the diagonal and the correlations between the constructs. This value is higher than other correlations between latent variables, indicating adequate discriminant validity of the measurements.

Table IV. Discriminant validity matrix (Fornell-Larcker Criterion)

Discriminant validity was assessed using the HTMT method (Henseler *et al.*, 2015), with a conservative 0.85 threshold. The HTMT ratio for group- and individual-focused transformational leadership was 0.832, below both 0.85 and 0.95 thresholds, confirming adequate discriminant validity.

Table V. Discriminant validity matrix (Heterotrait-Monotrait Ratio Criterion)

This provides good support for our claims of discriminant validity between our measures of transformational leadership at the group and individual level (Henseler *et al.*, 2015). On the other hand, the fit indices are the standardised root mean square residual (SRMR), which is 0.060, indicating a better model fit if the values are below 0.08. Lower values of d_ULS (1.086) and d_G (0.519) also indicate a better model fit. The chi-square (χ^2), which is 1291.401, compares the observed covariance matrix with the covariance matrix implied by the model.

Structural model analysis

The model shown in figure 2 has been elaborated from the reviewed literature and its analysis.

Figure 2. Structural model results (baseline model)

The PLS program generates T-statistics for significance testing using bootstrapping (Chin, 1998). In this process, 10,000 subsamples are taken from the original data with replacement, producing bootstrap standard errors and approximate T-values for structural path significance testing.

After bootstrapping, the R-squared values range from 0 to 1, with higher values indicating greater predictive capacity. R-squared values should be above 0.10 for the model to have adequate explanatory power, with significance at $t > 1.64$ (Fralk and Miller, 1992). Figure 3 and Table VI display the variance explained (R-squared) in the dependent constructs, confirming that the independent variables are adequate for the model.

Table VI. Structural model results

The standardised of the regression coefficients show the estimates of the relationships of the structural model, that is, the hypothesised relationships between constructs. So it will analyse the algebraic sign if there is change of sign, the magnitude and statistical significance is greater (T-student) of t (9999), one-tailed test) < 1.64 . After, the hypotheses were checked and validated and the relationships were positive, mostly with high significance (Table VII).

Table VII. Structural model results. Path significance using percentile bootstrap 95% confidence interval (n = 10.000 subsamples)

However, when the percentile bootstrap is applied to generate a 95% confidence interval using 10,000 resamples, all hypothesis are supported because their confidence interval does not include zero. Thus, these all hypotheses are adopted.

All of these results complete a basic analysis of PLS-SEM in our research. PLS-SEM result is shown in Figure 3.

Figure 3: Results of testing the model significance *p < 0.05. P < .01. * P < .001

Table VIII shows the amount of variance that each antecedent variable explains on each endogenous construct. The R² coefficient assesses the model's ability to explain and predict the variables of interest. These results are aligned when analysing the explanatory power of the model. R² figures are outstanding for almost all values, greater than 0.168. Thus, cross-validated redundancy measures show that the theoretical / structural model has a predictive relevance

Table VIII. Effects on endogenous variables (extended model)

Discussion

The analysis of artificial intelligence's impact on sustainable hotels reveals that generative artificial intelligence (GAI) significantly enhances green hotel processes, optimizing resource management and boosting employee happiness. The model's reliability and validity were confirmed, with strong relationships among variables supporting all proposed hypotheses.

Generative artificial intelligence improves green hotel operations by increasing efficiency (R² = 0.616), automating routine tasks, and personalizing customer service. This automation frees staff to engage in more meaningful and fulfilling activities, enhancing employee happiness and reducing stress. As one front-line employee noted, "With AI handling routine tasks, I can focus on guests' needs and feel more valued in my work environment." Another commented, "AI suggestions help me plan sustainable practices better, making my daily work less stressful and

more meaningful.” GAI also optimizes waste management and predictive maintenance, further improving operational efficiency and sustainability.

Employee awareness ($R^2 = 0.168$) and perceived sustainability ($R^2 = 0.227$) are influenced by GAI, promoting better resource management and greater environmental consciousness among staff. One employee stated, “AI reminds us of eco-friendly behaviors, encouraging me to take more initiative in sustainability efforts.” AI also supports staff training, monitors sustainability goals, and fosters eco-friendly behavior, aligning with findings by Carvalho and Ivanov (2023) and Demir and Demir (2023), who highlight AI’s role in augmenting employee environmental awareness. As another employee observed, “Training modules integrated with AI are interactive and practical, helping me understand sustainability without feeling overwhelmed.”

GAI positively impacts employee awareness (AG) (H1), explaining 16.80% of the variance, demonstrating that its implementation increases employees’ sustainability awareness (Carvalho & Ivanov, 2023; Demir & Demir, 2023). GAI also positively affects green hotel process management (MP) (H2), explaining 61.60% of the variance, indicating that AI optimizes internal operations and supports sustainable management (Demir & Demir, 2023; Dwivedi et al., 2023a). Additionally, GAI positively influences sustainability (SUS) (H3), accounting for 22.7% of the variance, showing that AI contributes to adopting greener and more responsible practices (Dwivedi et al., 2023a; Rathore, 2023).

Employee awareness significantly influences process management (H4) (24.29% of variance) and environmental performance (EP) (H5) (17.16% of variance), indicating that sustainability-conscious employees translate knowledge into improved operational and environmental practices. As one staff member remarked, “Being aware of sustainability goals through AI makes it easier to apply them in my daily work.” This aligns with prior research emphasizing the role of environmental awareness in fostering positive workplace behavior (Legrand et al., 2022; Majeed & Kim, 2022).

Sustainability (SUS) positively impacts environmental performance (EP) (H6) (9.30% of variance) and process management (H7) (12.33% of variance),

demonstrating that integrating sustainable practices strengthens operational efficiency (Legrand et al., 2022; Majeed & Kim, 2022; Chen et al., 2022; Sahoo et al., 2023). One employee noted, “When sustainability is part of our workflow, I feel proud of my contribution to both the hotel and the environment.” Process management (MP) directly influences environmental performance (EP) (H8) (18.93% of variance), showing that sustainability-oriented management improves environmental outcomes (Henche et al., 2020; Nisar et al., 2022).

Environmental performance (EP) significantly impacts employee happiness (EHA) (H9), explaining 46.5% of the variance, highlighting that employees experience higher job satisfaction when their work contributes to sustainability and environmental protection (Kamel et al., 2017; Rando Cueto et al., 2023; Ruiz-Rodríguez et al., 2023). An employee emphasized, “Knowing our efforts truly protect the environment gives me a strong sense of purpose and job satisfaction.”

Overall, the combined insights from PLS-SEM analysis and employee perspectives illustrate that GAI is not merely a technological tool but a catalyst for sustainability, improved management, and human-centered workplace well-being in green hotels. Employee testimonies underscore the practical impact of AI on daily operations, demonstrating how technology, sustainability, and human engagement converge to foster a productive, meaningful, and satisfying work environment.

Conclusions

This study validates a model examining the influence of generative artificial intelligence (GAI) and sustainability practices on green hotel management and their impact on employee happiness. The findings support three objectives. First, quantitative results show that environmental sustainability practices—such as resource optimization, waste reduction, and eco-friendly process management—not only improve environmental performance but also enhance job satisfaction and well-being. Second, the adoption of GAI improves operational efficiency by automating tasks, optimizing resources, and facilitating personalized guest services, which reduces employee stress and fosters workplace happiness. Third, advanced GAI technologies were found to strengthen employee awareness, sustainability perceptions, and process management, thereby improving environmental performance and explaining significant variance in employee happiness across eco-friendly hotels.

Theoretical Implications

This research contributes to literature on happiness management and green human resource management in hospitality. It extends the Resource-Based View (RBV) by showing that sustainability practices in hotel management act as intangible assets that generate ecological and competitive advantages while enhancing employee happiness. It also advances understanding of GAI adoption by demonstrating how technological capabilities and organizational competencies can simultaneously improve efficiency and employee well-being. Furthermore, it highlights the mediating role of process management and environmental performance in linking GAI technologies to employee happiness, underscoring their role in creating innovative, sustainable, and human-centered organizational cultures. These insights enrich theoretical frameworks that connect environmental responsibility with workplace happiness.

Practical Implications

For hotel managers and policymakers, the study offers actionable guidance. Embedding sustainability in daily operations not only supports ecological goals but also boosts employee morale, aligning workplace happiness with environmental responsibility. Effective GAI adoption requires leadership that treats digital tools as enablers of well-being rather than substitutes for human labor. Managers should design green HR policies that translate technological innovation into efficiency gains and happiness at work. Finally, advanced GAI applications—such as predictive maintenance, eco-monitoring, and interactive training—can increase environmental awareness and employees’ sense of purpose, reinforcing job satisfaction. By aligning sustainability, technology, and human capital, green hotels can achieve service excellence, competitive differentiation, and long-term value creation.

Limitations and Future Research

This study has limitations. The focus on Spanish eco-hotels restricts generalizability; cross-country studies could enhance validity. The emphasis on GAI and employee happiness overlooks multilevel dynamics; future research should integrate organizational, team, and individual perspectives. Expanding beyond hospitality may uncover cross-industry strategies to align GAI, sustainability, and workplace happiness.

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